Kappa Foundation

Group Assignment

The decision

It is 31 December 2022. Kappa Foundation has received USD 12 million in unrestricted cash. It funds education programmes and has made two commitments that cannot be cancelled. There are no other assets, liabilities or committed donations.

Commitment	Payment date	Amount
Scholarship programme	31 December 2024 (year 2)	USD 2.4 million
Learning centre	31 December 2027 (year 5)	USD 3.6 million

After meeting these payments, the trustees want the remaining fund at 31 December 2030 to have at least the purchasing power of USD 8 million in December 2022. This is a goal, not a promised investment return. The two scheduled payments are fixed nominal obligations.

The trustees are uncomfortable with investment losses approaching 15%, but they have not agreed how to measure that concern, over what period, or what compromises they would accept. They have not specified an inflation assumption, a return target, a risk-aversion parameter or a policy benchmark.

Eta, the treasurer, proposes using high-quality bonds to support the commitments and putting the balance into the portfolio with the strongest historical risk-adjusted performance. Theta, an external adviser, argues that factor exposures and persistent abnormal returns should determine the growth allocation. Both believe their approach offers a disciplined basis for investment.

Your group is the independent investment team. Recommend a policy and an initial allocation that the trustees could authorize. Decide which assumptions are defensible, what evidence supports your recommendation, and which promises the foundation cannot responsibly make.

The board's questions

- What should Kappa own, in what amounts, and under what investment policy?
- How credible is the plan for meeting the commitments and preserving the longer-term mission?
- What actually drives the proposed portfolio's return and risk? Which adviser's claims, if any, survive examination?
- What evidence would make you change your recommendation?

There is no prescribed optimal portfolio. A recommendation to change the trustees' goal is acceptable if supported by a feasible alternative and a quantified trade-off. You may not change the mandatory payments in the original case.

Exhibit 1: Available instruments

The foundation is tax-exempt, reports in USD, and may hold cash, the case bonds below, US common stocks and US-listed equity ETFs investing primarily in US equities. It may not borrow, sell short or use derivatives. Fractional holdings are permitted. There are no taxes or default losses on the case bonds. Propose and justify any additional portfolio limits and trading-cost assumptions.

The following are fictional executable quotes at the case decision date. Each bond has USD 1,000 face value. Coupons are paid annually on 31 December, starting in 2023; maturity includes the final coupon and principal. Settlement is immediately after the 2022 coupon date, so accrued interest is zero. The zero-coupon bonds have no intermediate payments. All five bonds are non-callable and default-free for this case.

Bond	Maturity from decision date	Annual coupon	Price per USD 1,000
Alpha	1 year	0.00%	958.77
Beta	2 years	0.00%	922.78
Gamma	3 years	4.00%	1,002.49
Delta	5 years	3.50%	986.07
Epsilon	8 years	4.50%	1,040.71

The exhibit specifies contractual cash flows and initial prices. It does not supply future resale prices, reinvestment rates or a complete spot curve. Any additional structure used in your analysis must be stated and defended.

Build a candidate equity universe of 8–12 US common stocks across at least four sectors, plus two such equity ETFs with meaningfully different investment mandates. Select your own tickers and explain their roles. You need not hold every candidate. Evidence of eligibility at the case date is required; present-day survival is not evidence that a security was an unbiased historical choice.

Select a simple, investable comparator for the growth allocation. A complicated optimized portfolio does not earn credit merely because it is complicated.

Exhibit 2: Information available to your team

The decision uses information dated no later than 31 December 2022. Use monthly market returns; choose and justify an estimation window with at least 60 common monthly observations ending in December 2022, drawn from January 2015–December 2022. January 2023–December 2025 is a separate evaluation period.

Retrieve the market data yourself in Python from Yahoo Finance, using yfinance, and retrieve the factor data yourself in Python from Kenneth French's website. Course workbook datasets, lecture caches and another group's processed returns are not substitutes for these downloads. Additional authoritative sources may support your assumptions, subject to the same information cutoff and source-documentation requirements.

The historical evaluation concerns the market-traded growth allocation. Evaluate the case bonds and contractual payments through their cash flows and explicit scenarios; do not fabricate historical quote series for these fictional bonds. A three-year growth evaluation is not evidence that the eight-year funding goal has been achieved.

This is a retrospective case. State the limits arising from security selection and revised historical data; do not present the exercise as a genuinely unseen live investment experiment.

Evidence supporting the decision

Organize the submission around your recommendation, not around the order of the lectures. The board requires the following evidence, but does not prescribe the objective function, the expected-return forecasts, the risk limits or the preferred model.

Area	Required evidence
Mandate and assumptions	A full USD 12 million allocation; a funded-payment plan; a defensible interpretation of risk tolerance and the real-value goal. Use a quantitative risk–return preference comparison and show how its assumed risk aversion affects the choice.
Bond and funding risk	Bond valuation and yield analysis; treatment of the incomplete term structure; duration and convexity estimates checked against exact repricing. Examine a parallel rate change and a change in curve shape, with magnitudes you justify. Relate the findings to the payment obligations.
Portfolio decision	One recommended allocation and one credible rejected alternative. Show how portfolio choice changes under sample-covariance and single-index risk estimates for the same candidate universe. Explain the expected-return inputs, constraints, implementation and sensitivity.
Market and factor explanations	Compare a market-only model with at least one multifactor model from Lecture 7 for the growth portfolio and at least two selected securities. Explain economic exposures, uncertainty and what any estimated alpha does—and does not—establish. Relate the results to expected returns and the investment decision.
Evidence against your recommendation	Compare the frozen growth strategy with its frozen benchmark in 2023–2025. Identify one plausible assumption change that reverses an important decision and quantify its threshold. Report inconvenient results as well as favourable ones.

A model result is not a decision by itself. Explain why the evidence is sufficient for the action you recommend, what remains uncertain, and what Kappa would monitor after implementation. No marks are awarded for high realized returns as such.

Data and reproducibility

- Keep the original downloaded files or untransformed download responses. Record their source, retrieval date, ticker or dataset identifier, requested dates, relevant download settings and package versions.
- Document return construction, units, timing, corporate-action treatment and exclusions. Make any differences between datasets visible rather than silently forcing them to agree.
- The submitted Python notebook or scripts must reproduce the analysis from the saved raw data without manual edits or internet access. Also include the functioning retrieval code used to obtain that data.
- Each material table or figure in the memo must be traceable to a named output generated by the code. Report failed downloads and how you resolved them; never fill missing observations with invented data.

Submission and review

Groups: 3–4 students. All times are Ho Chi Minh City, Vietnam time.

When	Submission or event
22 October 2026, 16:00	Initial decision record: at most two pages, candidate universe, estimation window, benchmark, initial dollar allocation, material assumptions and evaluation/rebalancing rule. Submit the working Python file and its dated data record.
23 October 2026, 16:00	Release of a revised mandate from the trustees. Retain the original submission and quantify the effect on your recommendation, using the same case decision date and information cutoff.
29 October 2026, 16:00	Final submission: committee memo, presentation slide deck, reproducible analysis and supporting data.

Freeze the initial policy and evaluation rule in the 22 October submission. Later changes must be recorded and compared with that original policy, rather than replacing it retrospectively. A predeclared rule may rebalance using information available at the time; future observations must not enter earlier decisions.

Submit one archive containing:

- A committee memo of no more than eight pages, including its tables and figures; references may occupy one additional page.
- The committee presentation slide deck, submitted as a PDF together with its editable source.
- The executed Python notebook or scripts, an environment/package file and concise instructions for rebuilding the results from the raw-data folder.
- Raw downloads, processed analysis data and generated outputs, with a source record.
- The 22 October submission and a short decision log recording consequential changes, rejected choices and the evidence behind them.
- A contribution statement identifying each member's substantive work and another member's check of it. Each member adds a short critique of the final recommendation.

Use of AI must comply with the course's academic-integrity policy. Disclose material AI assistance and the checks you performed. Every member is responsible for explaining the analysis submitted under their name.

Assessment	Marks
Mandate, assumptions and judgement	10
Bond valuation and funding risk	20
Data integrity and reproducibility	15
Portfolio construction and robustness	15
Market and factor interpretation	15
Evaluation and response to the mandate change	15
Committee memo and presentation	10
Total	100

Source entry points: Yahoo Finance (finance.yahoo.com); yfinance documentation (ranaroussi.github.io/yfinance); Kenneth French Data Library (mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html). Choose the relevant datasets and record the exact resources used.